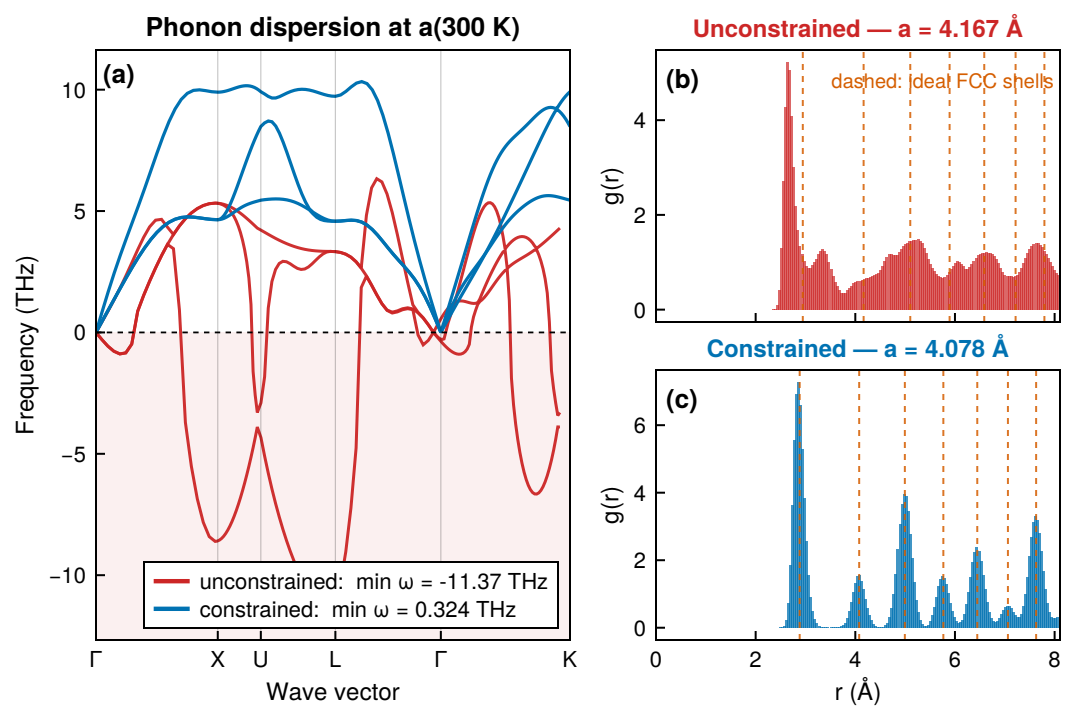
Phonon dispersion at a(300 K)
(a)
Frequency (THz)
10
5
0
−5
−10
Γ
X
U
L
Γ
K
Wave vector
unconstrained: min ω = -11.37 THz
constrained: min ω = 0.324 THz
Unconstrained — a = 4.167 Å
(b)
dashed: ideal FCC shells
g(r)
4
2
0
Constrained — a = 4.078 Å
(c)
g(r)
6
4
2
0
0
2
4
6
8
r (Å)